American Computer Science League

2022-2023 ● Contest 2: Short Problems Solutions ● Junior Division

1. Prefix/Infix/Postfix Notation

+ − * 2 3 / 4 2 ^ 2 3 = + − (* 2 3) (/ 4 2) (^ 2 3)
= + − (2 * 3) (4 / 2) (2 ^ 3)
= + (− 6 2) 8
= + (6 − 2) 8
= (+ 4 8)
= 4 + 8
= 12

C. 12

2. Prefix/Infix/Postfix Notation

$\frac{A+B}{C} - \frac{B*C}{A-C} = (A\ +\ B)\ /\ C\ -\ B\ *\ C\ /\ (A\ -\ C)$
$= (A\ B\ +)\ /\ C\ -\ (B\ C\ *)\ /\ (A\ C\ -)$
$= ((A\ B\ +)\ /\ C)\ -\ ((B\ C\ *)/(A\ C\ -))$
$= (A\ B\ +\ C\ /)\ -\ (B\ C\ *\ A\ C\ -\ /)$
$= A\ B\ +\ C\ /\ B\ C\ *\ A\ C\ -\ /\ -$

A. AB+C/BC*AC−/−

3. Bit-String Flicking

10100 OR 01100 AND 01011 = 10100 OR (01100 AND 01011)
= 10100 OR 01000
= 11100

B. 11100

4. Bit-String Flicking

(LCIRC-2 (RSHIFT-1 (NOT 01110))) = (LCIRC-2 (RSHIFT-1 10001))
= (LCIRC-2 01000)
= 00001

C. 00001

5. LISP

(SETQ M '(1 2 (5 1) (3 (4 5) 6) (2 (4))))
(CDR (CAR (CDR (REVERSE (CDR (CDR M))))))
= (CDR (CAR (CDR (REVERSE (CDR
(CDR '(1 2 (5 1) (3 (4 5) 6) (2 (4)))))))))
= (CDR (CAR (CDR (REVERSE (CDR '(2 (5 1) (3 (4 5) 6) (2 (4))))))))
= (CDR (CAR (CDR (REVERSE '((5 1) (3 (4 5) 6) (2 (4)))))))
= (CDR (CAR (CDR '((2 (4)) (3 (4 5) 6) (5 1)))))
= (CDR (CAR '((3 (4 5) 6) (5 1))))
= (CDR '(3 (4 5) 6))
= ((4 5) 6)

C. ((4 5) 6)

6. Prefix/Infix/Postfix Notation

6 3 / 2 2 ^ + 3 4 2 – * / = (6 3 /) (2 2 ^) + 3 (4 2 –) * /
= (6 / 3) (2 ^ 2) + 3 (4 – 2) * /
= (2 4 +) (3 2 *) /
= (2 + 4) (3 * 2) /
= (6 6 /)
= 6 / 6
= 1

B. 1

7. Prefix/Infix/Postfix Notation

4 6 + 7 5 – / 2 ^ 7 5 4 – * + = (4 6 +)(7 5 –) / 2 ^ 7 (5 4 –) * +
= ((+ 4 6)(– 7 5) /) 2 ^ (7 (– 5 4) *) +
= ((/ + 4 6 – 7 5) 2 ^) (* 7 – 5 4) +
= ((^ / + 4 6 – 7 5 2) (* 7 – 5 4) +)
= + ^ / + 4 6 – 7 5 2 * 7 – 5 4

E. +^/+46–752*7–54

8. Bit-String Flicking

(LSHIFT-2 (RCIRC-1 (NOT 10110))) = (LSHIFT-2 (RCIRC-1 01001))
= (LSHIFT-2 10100)
= 10000

B. 10000

9. Bit-String Flicking Let X = abcde LHS = (NOT 11011) AND X OR (RCIRC-1 01110) = (NOT 11011) AND abcde OR (RCIRC-1 01110) = (00100 AND abcde) OR 00111 = (00100 AND abcde) OR 00111 = 00c00 OR 00111 = 00111 RHS = 00111 LHS = RHS ⇒ 00111 = 00111 ⇒ a = *, b = *, c = *, d = *, e = * Therefore x = *****	D. *****
10. LISP (CDR (CAR (CDR '(b (c (a) b))))) = (CDR (CAR '((c (a) b)))) = (CDR '(c (a) b)) = ((a) b)	C. ((a) b)